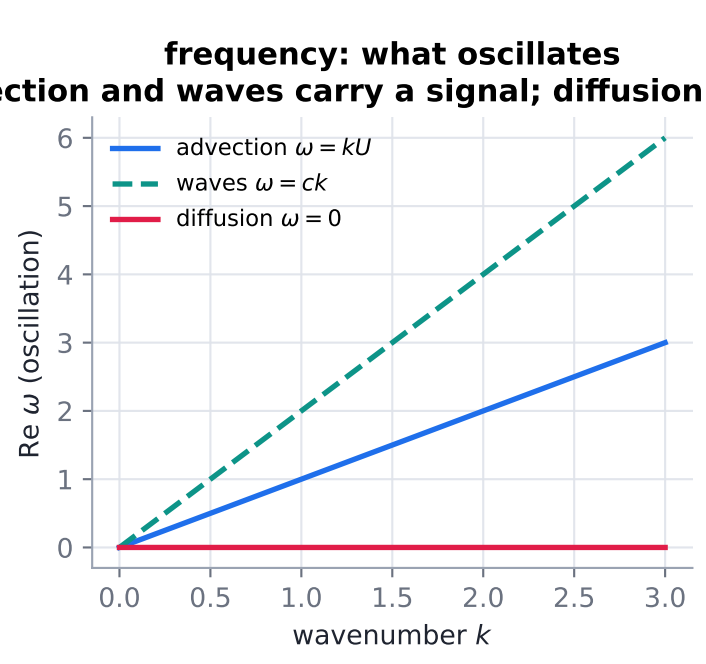
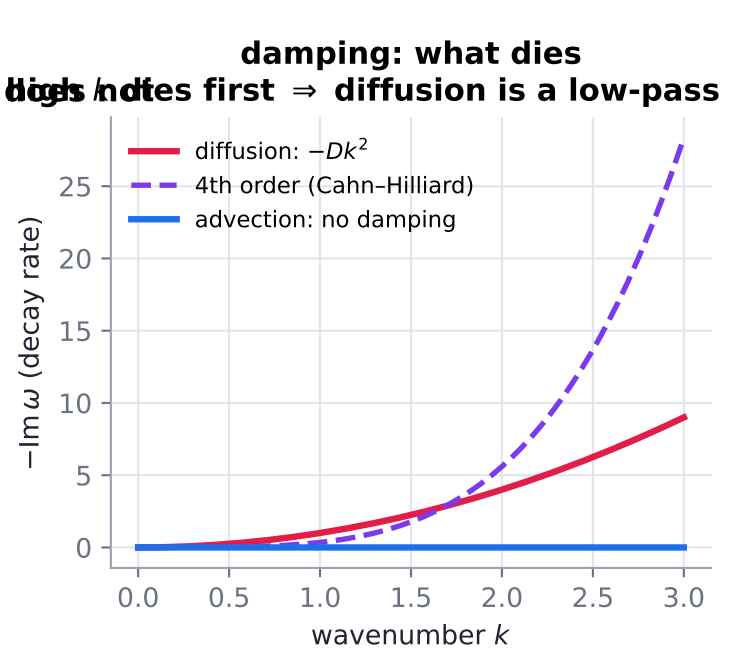
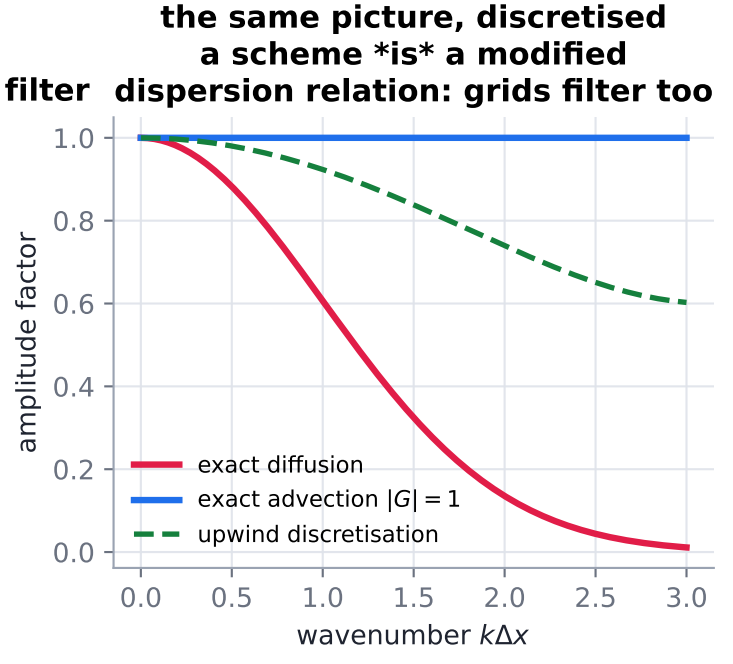

frequency: what oscillates
advection and waves carry a signal; diffusion does not
advection $\omega = kU$
waves $\omega = ck$
diffusion $\omega = 0$
Re ω (oscillation)
wavenumber k


damping: what dies
high k dies first ⇒ diffusion is a low-pass filter
diffusion: $-Dk^2$
4th order (Cahn–Hilliard)
advection: no damping
$-\mathrm{Im}\,\omega$ (decay rate)
wavenumber k


the same picture, discretised
a scheme *is* a modified dispersion relation: grids filter too
exact diffusion
exact advection $|G| = 1$
upwind discretisation
amplitude factor
wavenumber $k\Delta x$